EXPERIMENT-14
Smallest of 2 numbers.

AIM: To write an Assembly language program to find the smallest number in an array using 8085 Microprocessor in GNU Sim.

Software used: GNUSim8085.

Algorithm:

1. Initialize the count.
2. Get the input numbers
3. Compare the content of Accumulator(A) with HL pair for all input numbers.
4. Store the smallest number in the output register.
5. End the program.

PROGRAM:

```
LXI H, 8000H - load address into HL
MOV B, M - Copy memory to B.
INX H - Increment HL address
MOV A, M - copy memory to A.
CMP B - compare A and B.
JC STORE - Jump if A < B
MOV A, B - copy B to A
STA 8010H - Store result in memory
HLT - stop program.
```

Input

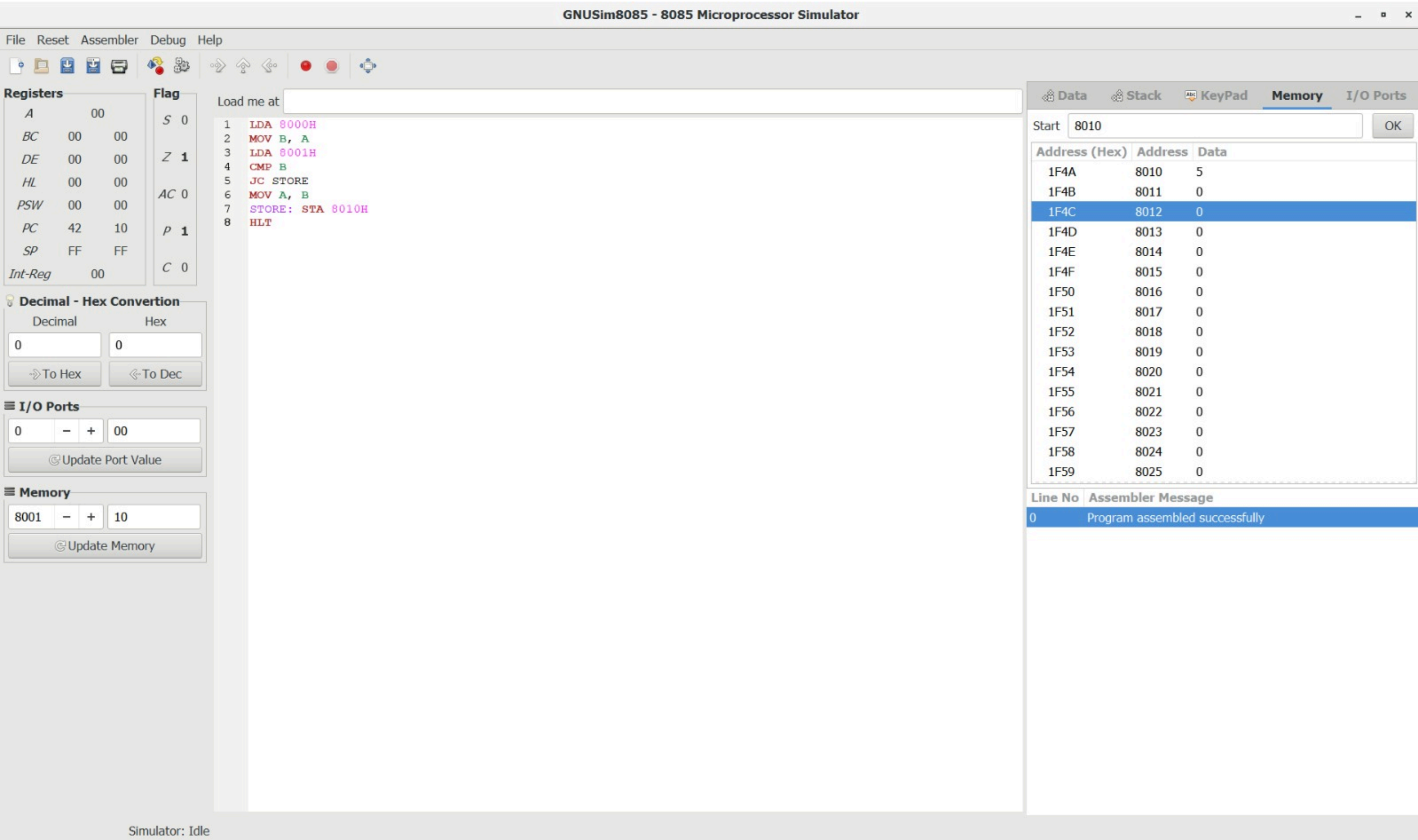
Address	Data
8000	10

Output:

Address	Data
8010	3

Result: Thus the assembly language program to find the smallest number in an array using 8085 Microprocessor in GNUSim is performed.

8/4


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 00
BC 00 00
DE 00 00
HL 00 00
PSW 00 00
PC 42 10
SP FF FF
Int-Reg 00
Flag
S 0
Z 1
AC 0
P 1
C 0
Load me at
LDA 8000H
MOV B, A
LDA 8001H
CMP B
JC STORE
MOV A, B
STORE: STA 8010H
HLT
Data Stack KeyPad Memory I/O Ports
Start 8010 OK
Address (Hex) Address Data
1F4A 8010 5
1F4B 8011 0
1F4C 8012 0
1F4D 8013 0
1F4E 8014 0
1F4F 8015 0
1F50 8016 0
1F51 8017 0
1F52 8018 0
1F53 8019 0
1F54 8020 0
1F55 8021 0
1F56 8022 0
1F57 8023 0
1F58 8024 0
1F59 8025 0
Line No Assembler Message
0 Program assembled successfully
Decimal - Hex Convertion
Decimal Hex
0 0
To Hex To Dec
I/O Ports
0 00
Update Port Value
Memory
8001 10
Update Memory
Simulator: Idle